LinkedIn Network Analytics Dashboard (Power BI)

Project Overview

In this project, LinkedIn exported data is analyzed to learn about trend and business communication behaviour and network structure.
Through Microsoft power BI, LinkedIn raw data were converted to three interactive dashboards that give information about:

- LinkedIn connection growth
- The composition of professional network.
- Top companies and job roles
- Patterns of messaging and communication.
- Most contacted connections

In this project, the aim is to show how individual LinkedIn information can be converted into valuable information through the application of data visualization and data analysis tools.

Why This Project Was Selected

LinkedIn provides valuable professional networking data, but it does not provide built-in analytics for users to analyze their network.

Most LinkedIn users cannot easily answer questions such as:

- How fast is my professional network growing?
- Which companies dominate my network?
- What job roles are common in my network?
- How active is my communication with connections?

This project was selected to convert **LinkedIn exported raw data into meaningful visual analytics using Power BI**.

Problem Statement

LinkedIn provides raw datasets but lacks analytical dashboards for understanding networking behavior.

Key problems addressed in this project:

1. No visualization of connection growth trends
2. Difficult to analyze professional network composition
3. No clear view of communication patterns
4. Hard to identify most contacted connections
5. Raw exported data is not structured for analysis

This project solves these issues by building **interactive dashboards that transform raw LinkedIn data into actionable insights**.

Tools & Technologies Used

LinkedIn Data Export	→Data Source
Microsoft Power BI	→Dashboard Development
Power Query	→Data Cleaning
DAX	→Calculations and Measures

Dataset Description

LinkedIn provides multiple datasets when exporting account data.
The following datasets were used in this project.

Connections Dataset

Contains details about LinkedIn connections.

Columns used:

- Company
- Position
- ConnectionDate
- ConnectionYear
- ConnectionMonth
- URL

Additional column created:

- **ConnectionMonthNumber**

Purpose:
Used to analyze:

- Network growth
- Top companies
- Job roles in the network

Invitations Dataset

Contains LinkedIn invitation activity.

Columns used:

- InviteName
- InviteDate
- InviteYear

- InviteMonth
- Direction

Additional column created:

- **InviteMonthNumber**

Purpose:

Used to analyze **invitation activity trends**.

Messages Dataset

Contains LinkedIn messaging activity.

Columns used:

- ConversationID
- Sender
- Receiver
- MessageDate
- MessageYear
- MessageMonth

Additional columns created:

- **MessageMonthNumber**
- **Message Type**

Purpose:
Used to analyze:

- Messaging activity
- Most contacted connections
- Sent vs received messages

Data Preparation

Data preparation was performed using **Power Query and Data Modeling in Power BI**.

Steps performed:

1. Imported LinkedIn datasets into Power BI.
2. Cleaned unnecessary columns.
3. Converted date fields into structured components.
4. Created Month and Year columns.
5. Created MonthNumber columns for proper sorting.
6. Created calculated columns using DAX.

Month Sorting Problem

When using month names in Power BI charts, months are sorted alphabetically instead of chronologically.

Example incorrect order:

April
August
December
February

This results in incorrect trend visualization.

Solution Implemented

To fix this issue, **MonthNumber columns were created**.

Examples:

Month	MonthNumber
January	1
February	2
March	3

MonthNumber Columns Created

ConnectionMonthNumber

Used to sort:

ConnectionMonth → Sort by → ConnectionMonthNumber

Ensures correct order for connection growth charts.

InviteMonthNumber

Used to sort:

InviteMonth → Sort by → InviteMonthNumber

Ensures correct order for invitation charts.

MessageMonthNumber

Used to sort:

MessageMonth → Sort by → MessageMonthNumber

Ensures correct order for messaging charts.

Calculated Column

Measures and Calculated Columns Used

In this project, several **DAX measures and calculated columns** were created to summarize LinkedIn data and support dashboard visualizations.

Total Connections

Total Connections = COUNTROWS(Connections)

Dataset Used

Connections

Purpose

Counts the total number of LinkedIn connections in the dataset.

Why this measure was used

The **COUNTROWS function** counts the number of records in the Connections table.
Each row represents one LinkedIn connection.

Where it is used in the dashboard

Used in:

Card Visual → "Total LinkedIn Connections"

This helps display the **overall size of the professional network**.

Total Invitations

Total Invitations = COUNTROWS(Invitations)

Dataset Used

Invitations

Purpose

Counts the total number of LinkedIn invitations.

Why this measure was used

Each row in the Invitations table represents one invitation activity.
Using **COUNTROWS** provides the total number of invitations.

Where it is used in the dashboard

Used in:

Card Visual → "Total Connection Invitations"

This helps measure **networking activity**.

Total Messages

Total Messages = COUNTROWS(messages)

Dataset Used

messages

Purpose

Counts the total number of messages exchanged in LinkedIn conversations.

Why this measure was used

Each row in the messages dataset represents one message.
Using **COUNTROWS(messages)** gives the total messaging volume.

Where it is used in the dashboard

Used in multiple visuals:

• Card → **Total Messages Exchanged**
• Line chart → **Monthly Messaging Activity**
• Column chart → **Yearly Messaging Activity**
• Stacked column chart → **Sent vs Received Messages**

Message Type (Calculated Column)

Message Type =
IF(messages[Sender] = "Gnaneshwar Sreepathi","Sent","Received")

Dataset Used

messages

Purpose

Classifies each message as either:

- Sent
- Received

Why this column was created

The dataset contains sender and receiver information but does not directly classify messages.

This calculated column allows us to:

• Compare sent vs received messages
• Analyze communication balance

Where it is used in the dashboard

Used in:

100% Stacked Column Chart

Messages Sent vs Received by Month

Fields used:

X-axis → MessageMonth
Y-axis → Total Messages
Legend → Message Type

This visualization shows **communication distribution between sent and received messages**.

Distinct Job Roles

Distinct Job Roles = DISTINCTCOUNT(Connections[Position])

Dataset Used

Connections

Purpose

Counts the number of **unique job roles** present in the LinkedIn network.

Why this measure was used

Different connections may have different job roles.
Using **DISTINCTCOUNT** helps measure **professional diversity**.

Where it is used in the dashboard

Used in:

Card Visual → Distinct Job Roles in My Network

This shows how many different professional roles exist within the network.

Total Companies

Total Companies = DISTINCTCOUNT(Connections[Company])

Dataset Used

Connections

Purpose

Counts the number of **unique companies represented in the LinkedIn network**.

Why this measure was used

Many connections may work for the same company.
Using **DISTINCTCOUNT** helps identify **company diversity** in the network.

Where it is used in the dashboard

Used in:

Card Visual → Companies Represented in My Network

This helps understand the **organizational diversity of the professional network**.

Why COUNTROWS Was Used Instead of COUNT

Using **COUNTROWS()** is considered a best practice in Power BI when counting records in a table.

Advantages:

• Counts all rows in the dataset
• Avoids issues if a specific column contains blank values
• More reliable when working with entire tables

Example:

COUNTROWS(messages)

Counts all rows in the messages table.

Summary of Measures

Measure	Purpose
Total Connections	→Counts total LinkedIn connections
Total Invitations	→Counts invitation activity
Total Messages	→Counts total messages exchanged
Message Type	→Classifies messages as Sent or Received
Distinct Job Roles	→Counts unique job roles
Total Companies	→Counts unique companies in the network

Dashboard Structure

The project contains **Four dashboards**:

1. Dashboard Summary (Landing Page)
2. LinkedIn Profile Analytics Overview
3. LinkedIn Network Insights
4. LinkedIn Communication Insights

1. **Dashboard Landing Page** (Summary Page)

A **landing page** was created to act as the entry point for the Power BI report.

The purpose of this page is to provide users with an overview of the project and allow them to easily navigate to different dashboards.

Components of the Summary Page

The summary page includes:

- Dashboard title
- Project description
- Author branding
- LinkedIn theme design
- Navigation buttons for each dashboard

Navigation Buttons

Three navigation buttons were implemented:

- **Profile Analytics**
- **Network Insights**
- **Communication Insights**

Each button is configured using **Page Navigation actions** to redirect users to the respective dashboard pages.

Purpose

This landing page improves:

- User navigation
- Dashboard presentation
- Professional appearance of the report

2. Personal Branding

The dashboard includes **personal branding elements** to identify the project author.

Branding Elements Added

- Author photo
- Author name
- LinkedIn theme design

Example displayed on dashboard:

Author: Gnaneshwar Sreepathi

Purpose

Adding branding elements helps:

- Identify the project creator
- Make the dashboard suitable for portfolio demonstration
- Improve presentation quality when sharing the dashboard

3. LinkedIn Themed Design

Since the dataset is derived from **LinkedIn data export**, the dashboard design follows a **LinkedIn-inspired theme**.

Design Elements Used

- LinkedIn logo
- Blue color theme
- Professional background design
- Clean layout structure

Purpose

Using a consistent theme improves:

- Visual storytelling

- Professional appearance
- Brand association with LinkedIn data

The dashboard includes a dedicated landing page to improve navigation and user experience.

Dashboard 2: LinkedIn Profile Analytics Overview

Purpose

Provides a high-level summary of networking activity and messaging trends.

Chart 1 — Total LinkedIn Connections

Chart Type: **Card**

Dataset Used: Connections

Measure Used: Total Connections

Purpose:

Shows total number of LinkedIn connections.

Value:

Helps understand the size of the professional network.

Chart 2 — Total Connection Invitations

Chart Type: **Card**

Dataset Used: Invitations

Measure Used: Total Invitations

Purpose:

Displays networking activity through invitations.

Chart 3 — Total Messages Exchanged

Chart Type: **Card**

Dataset Used: Messages

Measure Used: Total Messages

Purpose:

Shows total messaging activity.

Chart 4 — Monthly Connection Growth by Year

Chart Type: **Line Chart**

Dataset Used: Connections

X-Axis: ConnectionMonth

Y-Axis: Count of ConnectionDate

Legend: ConnectionYear

Purpose:

Shows connection growth trends across months and years.

Insight:

Helps identify periods of rapid network expansion.

Chart 5 — Top Companies in My Network

Chart Type: **Bar Chart**

Dataset Used: Connections

X-Axis: Count of Connections

Y-Axis: Company

Purpose:

Identifies companies most represented in the network.

Chart 6 — Monthly Connection Invitations Sent

Chart Type: **Column Chart**

Dataset Used: Invitations

X-Axis: InviteMonth

Y-Axis: Count of InviteName

Purpose:

Shows invitation activity across months.

Chart 7 — Monthly Messaging Activity

Chart Type: **Line Chart**

Dataset Used: Messages

X-Axis: MessageMonth

Y-Axis: Total Messages

Purpose:

Shows monthly communication trends.

Chart 8 — Filter by connection year

Chart Type: **Slicer**

Dataset Used: Connection

Field: connectionyear

Also created

Chart Focus using Bookmarks

One advanced feature implemented in this project is **chart focus using Power BI bookmarks**.

This functionality allows users to view a specific visualization in a larger and more detailed format.

Implementation Steps

The following steps were performed to implement this feature:

1. The original chart was duplicated.
2. The duplicate chart was resized to create a **larger detailed view**.
3. The large chart was initially **hidden from the dashboard view**.
4. A **bookmark was created to display the detailed chart view**.
5. Another **bookmark was created to return to the original dashboard layout**.
6. Buttons were added to trigger the bookmarks.

Bookmarks Created

Two bookmarks were created for this interaction:

- **Bookmark_ShowTopCompanies**
  Displays the enlarged view of the chart.
- **Bookmark_Back**
  Returns the user to the original dashboard layout.

Buttons Used

Two buttons were created to control the interaction:

- **View Details** – Opens the enlarged visualization.
- **Back** – Returns to the dashboard view.

Purpose of this Feature

This feature allows users to:

- Focus on a specific visualization
- Explore chart details more clearly
- Maintain a clean dashboard layout without overcrowding visuals

Benefit

Using bookmarks improves the **user interaction and navigation experience**, making the dashboard behave more like an interactive analytical application rather than a static report.

Dashboard 3: LinkedIn Network Insights

Purpose

Analyzes professional diversity in the LinkedIn network.

Chart 1 — Companies Represented in My Network

Chart Type: **Card**

Measure: Total Companies

Purpose:

Displays number of unique companies.

Chart 2 — Distinct Job Roles

Chart Type: **Card**

Measure: Distinct Job Roles

Purpose:

Shows diversity of job roles.

Chart 3 — Top Job Roles Across My Network

Chart Type: **Bar Chart**

Dataset Used: Connections

X-Axis: Count of FullName / Connections

Y-Axis: Position

Purpose:

Identifies most common job roles.

Chart 4 — Top Companies Represented in My Network

Chart Type: **Bar Chart**

Dataset Used: Connections

X-Axis: Count of Connections

Y-Axis: Company

Purpose:

Shows companies with most connections.

Chart 5 — LinkedIn Network Growth Over Time

Chart Type: **Column Chart**

Dataset Used: Connections

X-Axis: ConnectionYear

Y-Axis: Count of Connections

Purpose:

Shows yearly network growth.

Chart 6 — My Professional Career Journey

Chart Type: **Bar Chart**

Dataset Used: Positions

X-Axis: StartDate

Y-Axis: Company

Purpose:

Visualizes career progression.

Chart 7 — My Educational Background

Chart Type: **Bar Chart**

Dataset Used: Education

X-Axis: Count of Degree

Y-Axis: University

Purpose:

Shows educational institutions attended.

Chart 8 — Filter by Job role

Chart Type: **Slicer**

Dataset Used: Connection

Field: position

Chart 9 — Filter by Company

Chart Type: **Slicer**

Dataset Used: Connection

Field: company

Also added

Drill Down Functionality

Drill-down capability allows users to explore data at multiple levels of granularity.

For example:

LinkedIn Network Growth Over Time

- X-Axis Level 1 → ConnectionYear
- X-Axis Level 2 → ConnectionMonth

Users can:

- View yearly growth trends
- Drill down into monthly connection growth within a specific year

This helps in identifying **detailed networking trends over time**.

Custom Tooltips

Custom tooltips were implemented to provide **additional context without cluttering the main dashboard**.

Example:

Career Journey Visual

When users hover over the visual, additional details are displayed:

- Company name
- Job title
- Start date
- End date

This improves the **data storytelling capability of the dashboard**.

Report Page Tooltip

A dedicated tooltip page was created to display **company-level insights**.

Tooltip displays:

- Company name
- Number of connections in that company
- Number of job roles associated with that company

This tooltip appears when users hover over company visuals.

Benefits:

- Provides detailed information instantly
- Keeps the main dashboard clean and minimal
- Enhances interactivity

Dashboard 4: LinkedIn Communication Insights

Purpose

Analyzes communication patterns with LinkedIn connections.

Chart 1 — Total Messages Exchanged

Chart Type: **Card**

Measure: Total Messages

Chart 2 — Total Conversations

Chart Type: **Card**

Measure: Total Conversations

Chart 3 — Monthly Messaging Activity

Chart Type: **Line Chart**

Dataset Used: Messages

X-Axis: MessageMonth

Y-Axis: Total Messages

Purpose:

Shows monthly messaging trends.

Chart 4 — Most Contacted Connections

Chart Type: **Bar Chart**

Dataset Used: Messages

X-Axis: Total Messages

Y-Axis: Receiver

Purpose:

Identifies most contacted connections.

Chart 5 — Yearly Messaging Activity

Chart Type: **Column Chart**

Dataset Used: Messages

X-Axis: MessageYear

Y-Axis: Total Messages

Purpose:

Shows messaging growth across years.

Chart 6 — Messages Sent vs Received

Chart Type: **100% Stacked Column Chart**

Dataset Used: Messages

X-Axis: MessageMonth

Y-Axis: Total Messages

Legend: Message Type (Sent / Received)

Purpose:

Compares sent vs received messages across months.

Chart 7 — Filter by Year

Chart Type: **Slicer**

Dataset Used: Messages

Field: Message year

Chart 8 — Filter by Contact

Chart Type: **Slicer**

Dataset Used: Messages

Field: Receiver

Key Insights

Major insights derived from the analysis include:

• **LinkedIn network growth accelerated significantly after 2023**, indicating increased professional networking activity during recent years.

• **Certain companies dominate the network**, with organizations such as *Infosys* and *Tata Consultancy Services (TCS)* representing a large portion of the connections.

• **Technology-related roles are most common**, particularly positions such as *Data Analyst, System Engineer, Software Engineer, and Technology Analyst*.

• **Messaging activity increased steadily over time**, suggesting stronger engagement and communication within the professional network.

• **A small group of connections accounts for a significant portion of messaging activity**, indicating stronger communication relationships with specific contacts.

"Value of the Dashboard"

Value of the Dashboard

This dashboard provides meaningful insights into professional networking patterns, including:

• **Tracking professional network growth over time**

• **Understanding company representation within the network**

• **Analyzing the diversity of job roles among connections**

• **Monitoring messaging activity and communication trends**

• **Identifying key professional relationships based on interaction frequency**

The dashboard enables users to **explore networking trends, analyze professional relationships, and better understand engagement patterns within their LinkedIn network through interactive visualizations.**

Advanced Dashboard Features

The dashboard incorporates several advanced Power BI capabilities to enhance user interaction and analytical exploration:

• Interactive slicers for dynamic filtering
• Drill-down functionality for exploring hierarchical data

• Bookmark-based navigation for focused chart views
• Custom report page tooltips for additional context
• Interactive navigation between dashboard pages

These features improve the **user experience and allow deeper exploration of networking insights**.

Data Source

LinkedIn Data Export

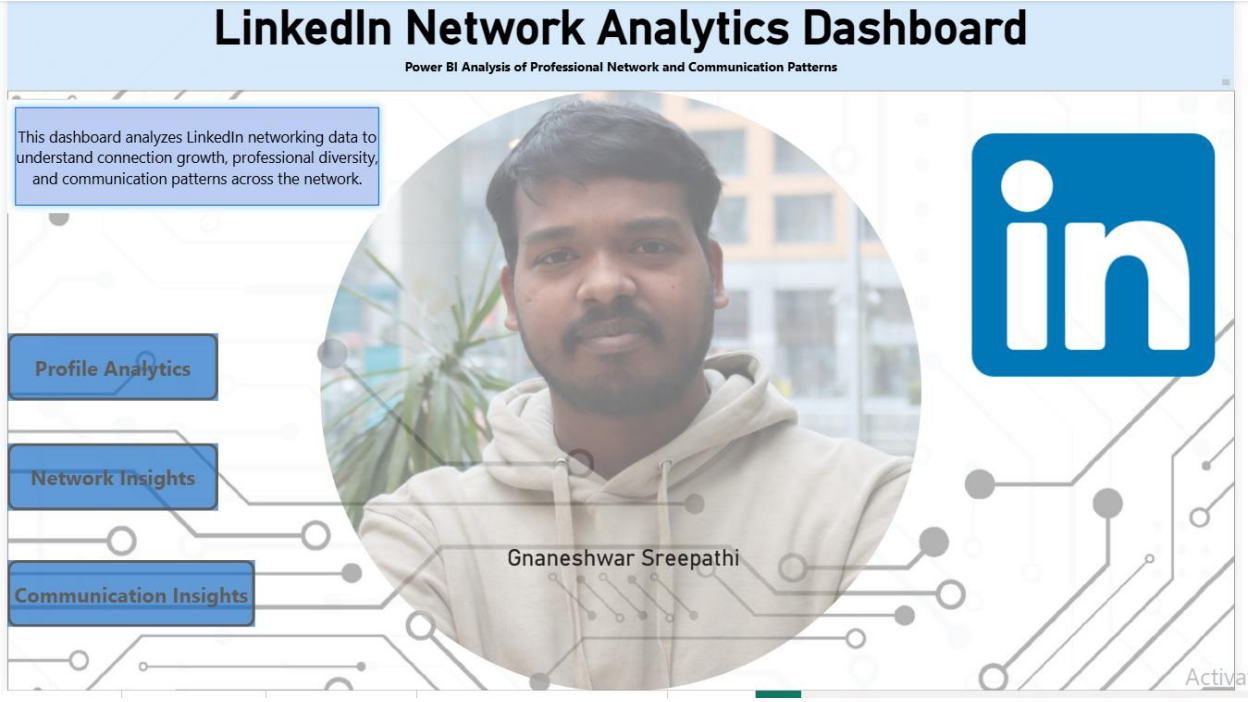
Visualization Tool

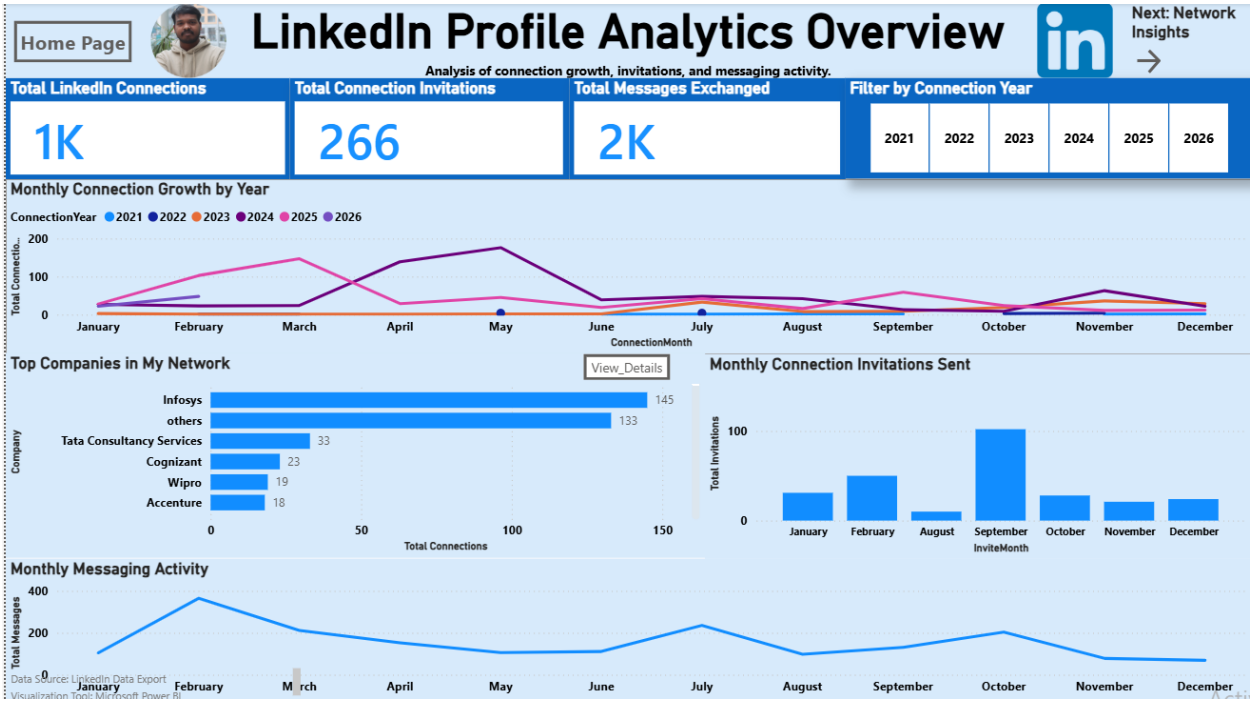
Microsoft Power BI

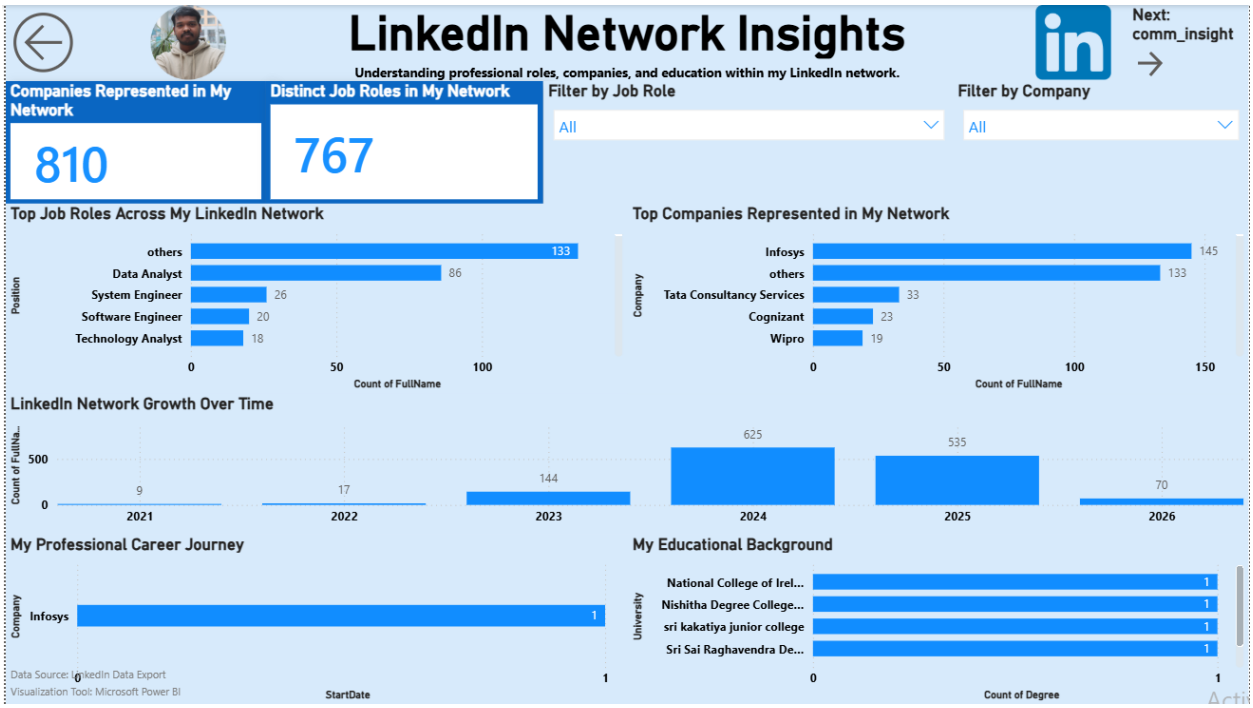
Repository Structure

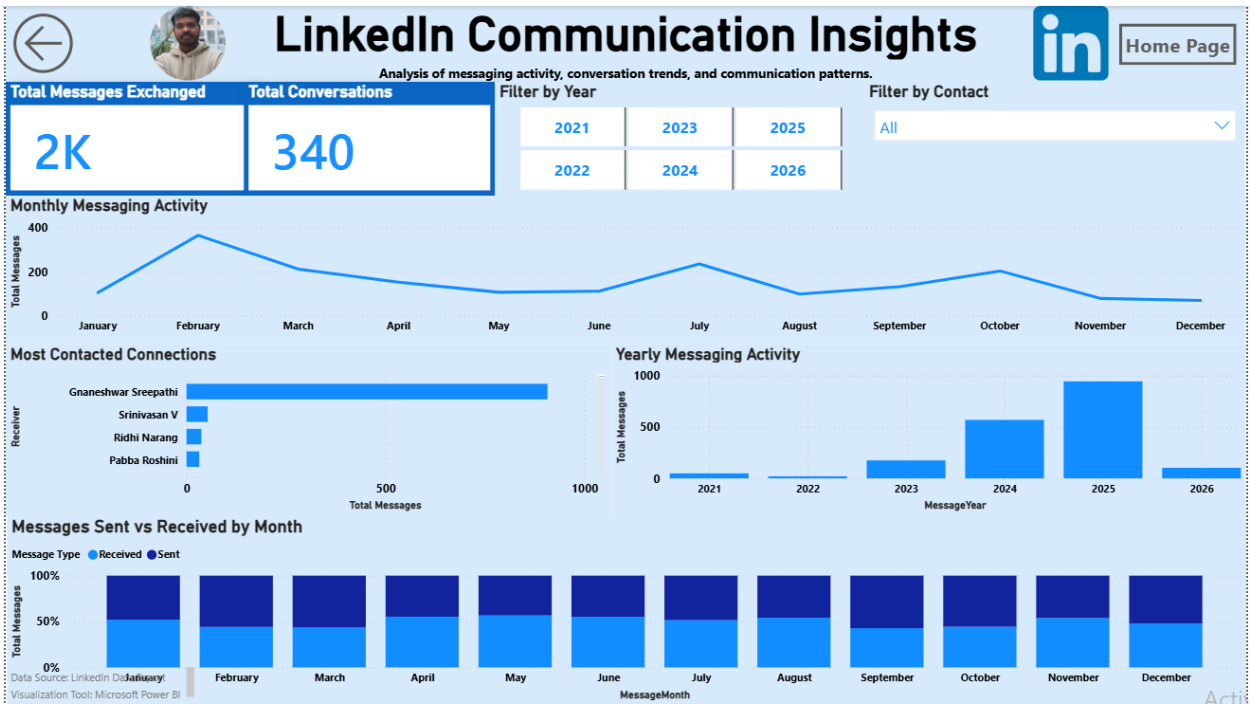
```
linkedin-network-analytics
│
├── dataset
│   ├── connections.csv
│   ├── invitations.csv
│   ├── messages.csv
│   ├── education.csv
│   ├── positions.csv
│
├── dashboards
│   └── linkedin_network_dashboard.pbix
│
├── screenshots
│   ├── dashboard_summary.png
│   ├── profile_overview.png
│   ├── network_insights.png
│   ├── communication_insights.png
│
├── documentation
```

```
│   └── project_documentation.md
│
└── README.md
```

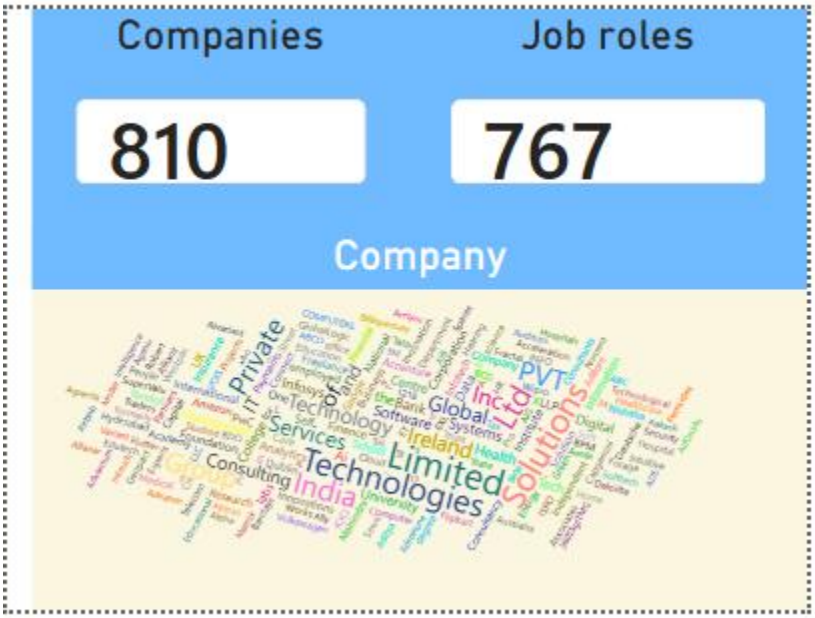





LinkedIn Network Insights
Understanding professional roles, companies, and education within my LinkedIn network.
Next: comm_insight
Companies Represented in My Network
810
Distinct Job Roles in My Network
767
Filter by Job Role
All
Filter by Company
All
Top Job Roles Across My LinkedIn Network
Position
others 133
Data Analyst 86
System Engineer 26
Software Engineer 20
Technology Analyst 18
0 50 100
Count of FullName
Top Companies Represented in My Network
Company
Infosys 145
others 133
Tata Consultancy Services 33
Cognizant 23
Wipro 19
0 50 100 150
Count of FullName
LinkedIn Network Growth Over Time
Count of FullNa...
500
0
9 17 144 625 535 70
2021 2022 2023 2024 2025 2026
My Professional Career Journey
Company
Infosys 1
0 1
StartDate
My Educational Background
University
National College of Irel... 1
Nishitha Degree College... 1
sri kakatiya junior college 1
Sri Sai Raghavendra De... 1
0 1
Count of Degree
Data Source: LinkedIn Data Export
Visualization Tool: Microsoft Power BI


LinkedIn Communication Insights
Analysis of messaging activity, conversation trends, and communication patterns.
Home Page
Total Messages Exchanged
2K
Total Conversations
340
Filter by Year
2021 2023 2025
2022 2024 2026
Filter by Contact
All
Monthly Messaging Activity
Total Messages
400
200
0
January February March April May June July August September October November December
Most Contacted Connections
Receiver
Gnaneshwar Sreepathi
Srinivasan V
Ridhi Narang
Pabba Roshini
0 500 1000
Total Messages
Yearly Messaging Activity
Total Messages
1000
500
0
2021 2022 2023 2024 2025 2026
MessageYear
Messages Sent vs Received by Month
Message Type Received Sent
Total Messages
100%
50%
0%
January February March April May June July August September October November December
MessageMonth
Data Source: LinkedIn Data Export
Visualization Tool: Microsoft Power BI

Tooltip


Companies
Job roles
810
767
Company
Private
IT
PVT
Inc
Ltd
Solutions
Global
Technology
Services
Ireland
Limited
Technologies
India
Consulting